

goalv0a — the centre of the even hybrid family quantum GL_1

Signed July 19, 2026; this is the notation-migrated edition of August 10–11, 2026. The statement is unchanged — only symbols were renamed, by the Sandbox-wide dictionary $Q \mapsto q$, $q \mapsto v$, $z \mapsto t$: the relation formerly written $Q = q^2$ now reads $q = v^2$, and the family parameter formerly written z is now written t . (In the current Sandbox-wide frame, v and z denote the square roots $v^2 = q$ and $z^2 = t$; only v and t occur below.)

Let $G = GL_1$: the root system is $\Phi = \emptyset$, the Weyl group is $W = \{1\}$, the cocharacter lattice is $\Lambda = \mathbb{Z}$, and $X_\Lambda = \mathbb{Z}$. Let

$$A_t = \mathbb{Z}[v^{\pm 1}, t^{\pm 1}], \quad F_t = \mathbb{Q}(v)[t^{\pm 1}], \quad q = v^2,$$

with v, t independent indeterminates.

Definition 1 (Hybrid family quantum GL_1). *U is the F_t -algebra with invertible pairwise commuting generators $K_{n,+}, K_{n,-}$ ($n \in \mathbb{Z}$) subject to*

$$K_{0,\pm} = 1, \quad K_{m+n,\pm} = K_{m,\pm} K_{n,\pm}, \quad K_{n,+} K_{n,-} = t^n.$$

There are no generators e_i, F_i , since $\Phi = \emptyset$. Writing $T^\pm := K_{1,\pm}$, the monomials $(T^+)^w$ ($w \in \mathbb{Z}$) form a F_t -basis of U .

Definition 2 (Even Newton lattice). *A toral element is an element of $F_t[(T^+)^{\pm 1}]$; it is even when all its toral exponents are even — equivalently, it is a Laurent polynomial in $\check{Y} := (T^+)^2$. For $\chi \in X_\Lambda$ let ev_χ be the A_t -linear map on even toral elements determined by $\text{ev}_\chi(\check{Y}) = q^\chi$. The even Newton lattice is the A_t -submodule*

$$\mathcal{N}^{\text{ev}} = \{ f \in F_t[\check{Y}^{\pm 1}] : \text{ev}_\chi(f) \in A_t \text{ for every } \chi \in \mathbb{Z} \}.$$

Definition 3 (Even hybrid family integral form). *Since $\Phi^+ = \emptyset$, the even hybrid family integral form is the A_t -submodule*

$$U^{\text{hyb, ev}} = \mathcal{N}^{\text{ev}} \subset U,$$

and its central part is $Z(U^{\text{hyb, ev}}) = U^{\text{hyb, ev}} \cap Z(U \otimes_{F_t} \text{Frac}(A_t))$.

Definition 4 (Harish–Chandra projection; shifted Weyl action). *Every element of U is toral, so the Harish–Chandra projection HC is the identity map on U . With*

$$Y_n = t^{-n} K_{n,+}^2 \quad (n \in \mathbb{Z}),$$

every even toral element is a Laurent polynomial in the Y_n , and the shifted Weyl action of W is the F_t -algebra action $w(Y_n) = Y_{wn}$; set $(\mathcal{N}^{\text{ev}})^W = \{f \in \mathcal{N}^{\text{ev}} : wf = f \text{ for all } w \in W\}$.

Definition 5 (Newton divided classes). *For $r \geq 0$ set*

$$\nu_r(X) = \prod_{s=0}^{r-1} \frac{X - q^s}{q^r - q^s} \in \mathbb{Q}(v)[X], \quad \nu_0(X) = 1.$$

Goal. For $G = GL_1$, with $\Lambda = \mathbb{Z}$:

(i) $U^{\text{hyb, ev}}$ is an A_t -subalgebra of U ;

(ii) the Harish-Chandra projection restricts to an isomorphism of A_t -algebras

$$\text{HC} : Z(U^{\text{hyb, ev}}) \xrightarrow{\sim} (\mathcal{N}^{\text{ev}})^W;$$

$$(iii) \mathcal{N}^{\text{ev}} = \sum_{u \in \mathbb{Z}} \sum_{r \geq 0} A_t \check{Y}^u \nu_r(\check{Y}).$$